CS5234 Mock Quiz 2 — Solutions

Lectures 5, 7–10 · open book · 50 minutes · 20 points

Questions 1–4: one correct option each (1 point). Questions 5–6: show your reasoning; an essay question left completely blank receives 1 mark. log denotes the natural log unless stated.

1 [1 pt]

A stream of updates (i, Δ) with arbitrary integer Δ (frequencies may become negative). You want, for every i, an estimate with $|\tilde{x}_i - x_i| \le \varepsilon \|x\|_2$. Which sketch should you use?

(A) Misra-Gries

(B) Count-Min

(C) Count Sketch

(D) KMV

Solution. **(C)**. Misra-Gries and KMV are insert-only; Count-Min needs $x_i \ge 0$ (strict turnstile) and gives an ℓ_1 guarantee. Count Sketch is a linear sketch with random signs: unbiased, variance $\le \|x\|_2^2/k$, works in any turnstile stream.

2 [1 pt]

In the KMV distinct-elements algorithm (output tm/X where X is the t-th smallest hash value), what is the weakest independence of the hash function that the lecture's analysis needs?

(A) Fully (mutually) independent

(B) 4-wise independent

(C) Pairwise independent

(D) No independence is needed

Solution. **(C)**. The analysis counts $Y = \sum Y_i$ (hash values below a threshold) and only uses $\mathrm{Var}Y = \sum \mathrm{Var}Y_i$ plus Chebyshev. Variances add as soon as the Y_i are pairwise independent, so a pairwise-independent family ($O(\log m)$ bits) suffices.

3 [1 pt]

The Morris counter increments C with probability 2^{-C} on each 1 and outputs $2^C - 1$. Let A be the number of 1s. Which statement is true?

(A) $\mathbb{E}[2^C] = A$.

(B) $\mathrm{Var}(2^C) = \frac{A(A-1)}{2}$, so averaging $\frac{10}{\varepsilon^2}$ independent copies gives a $(1 \pm \varepsilon)$-estimate with probability ≥ 0.9.

(C) Each copy needs $\Theta(\log n)$ bits, the same as an exact counter.

(D) The analysis needs a pairwise-independent hash function.

Solution. **(B)**. With $X_i = 2^{C_i}$: $\mathbb{E}[X_{i+1} \mid X_i = x] = x + 1$ so $\mathbb{E}[2^C] = A + 1$ ((A) is off by one; that is why the output subtracts 1). $\mathbb{E}[X_{i+1}^2 \mid x] = x^2 + 3x$ gives $\mathrm{Var}X_A = \frac{A(A-1)}{2} < A^2$. The mean of $k = \frac{10}{\varepsilon^2}$ copies has variance $< \frac{\varepsilon^2 A^2}{10}$, so Chebyshev gives failure $< \frac{1}{10}$. (C) is wrong: $C \approx \log A$, so a copy stores $O(\log \log n)$ bits. (D) is wrong: Morris uses fresh coin flips, no hashing.

4 [1 pt]

In an edge stream, the algorithm keeps H and adds an arriving edge iff it does not create a cycle of length ≤ 4 in H (i.e. $k = 2$). Which is true?

(A) H is a 3-spanner of G with $O(n^{3/2})$ edges.

(B) H is a 4-spanner of G with $O(n\sqrt{n}\log n)$ edges.

(C) H is a 3-spanner, but it can have $\Theta(n^2)$ edges.

(D) H preserves all distances exactly.

Solution. **(A)**. A rejected edge (u, v) would close a cycle of length $\leq 2k = 4$, so H already has a u–v path of length $\leq 3 = 2k - 1$; replacing every edge of a shortest path gives stretch 3. H has girth > 4, so by the BFS / min-degree argument it has $O(n^{1+1/k}) = O(n^{3/2})$ edges, which rules out (C). The stretch is $2k - 1 = 3$, not 4, and distances are not exact (a rejected edge's endpoints are at distance up to 3).

5 [9 pts]

Two vectors $x, y \in \mathbb{Z}^m$ start at 0. A single stream interleaves updates (x, i, Δ) meaning $x_i \mathrel{+}= \Delta$ and (y, i, Δ) meaning $y_i \mathrel{+}= \Delta$, with arbitrary integer Δ (turnstile). All entries stay bounded by poly(n). Design a streaming algorithm that outputs $D = \|x - y\|_2^2$ within $(1 \pm \varepsilon)$ with probability $1 - \delta$.

(a, 2 pts) Describe one basic estimator: what is stored and how each update is processed. Why must x and y use the *same* random signs?

Solution. Pick a 4-wise independent $p : [m] \to \{\pm 1\}$, keep one counter Z. On (x, i, Δ): $Z \mathrel{+}= p_i\Delta$; on (y, i, Δ): $Z \mathrel{-}= p_i\Delta$. Then $Z = \sum_i p_i x_i - \sum_i p_i y_i = \sum_i p_i z_i$ with $z = x - y$; output Z^2. This is AMS on z, which works because the sketch is linear. With different signs q for y we would get $\sum(p_i x_i - q_i y_i)$, which is not a sketch of z and does not estimate $\|z\|_2^2$. *[1: correct sketch/update; 1: linearity/same signs]*

(b, 4 pts) Show $\mathbb{E}[Z^2] = D$ and $\mathrm{Var}[Z^2] \leq 2D^2$. State exactly where you use 4-wise independence.

Solution. $\mathbb{E}Z^2 = \sum_i z_i^2 \mathbb{E}[p_i^2] + \sum_{i \neq j} z_i z_j \mathbb{E}[p_i p_j] = \sum z_i^2 = D$, since $p_i^2 = 1$ and $\mathbb{E}[p_i p_j] = \mathbb{E}p_i \mathbb{E}p_j = 0$ (pairwise). For the 4th moment, $\mathbb{E}Z^4 = \sum_{i,j,k,l} z_i z_j z_k z_l \mathbb{E}[p_i p_j p_k p_l]$; by 4-wise independence a term survives only if indices pair up: $i = j = k = l$ (gives $\sum z_i^4$) or two distinct pairs ($\binom{4}{2}/2 \cdot 2 = 6$ orderings per unordered pair $\{i, j\}$, gives $6\sum_{i<j} z_i^2 z_j^2$). Since $D^2 = \sum z_i^4 + 2\sum_{i<j} z_i^2 z_j^2$, $\mathrm{Var}Z^2 = \mathbb{E}Z^4 - D^2 = 4\sum_{i<j} z_i^2 z_j^2 \leq 2D^2$. *[1: expectation; 1: cross terms vanish by independence; 1: 4th moment count; 1: final bound]*

(c, 3 pts) Choose the number of copies and give the total space in bits.

Solution. Average $k = 6/\varepsilon^2$ independent copies: $\mathrm{Var}(\bar{Z}) \leq 2D^2/k$, Chebyshev $\Pr[|\bar{Z} - D| \geq \varepsilon D] \leq \frac{2D^2}{k\varepsilon^2 D^2} = \frac{1}{3}$. Take the median of $r = O(\log \frac{1}{\delta})$ such averages; by Chernoff the median fails only if half the groups fail, probability $\leq \delta$. Each copy stores a counter of $O(\log n)$ bits (entries are poly(n), so $|Z| \leq m \cdot \mathrm{poly}(n)$: $O(\log m + \log n)$ bits) and a 4-wise hash seed of $O(\log m)$ bits. Total $O(\varepsilon^{-2} \log \frac{1}{\delta}(\log m + \log n))$. *[1: mean trick + Chebyshev; 1: median trick; 1: space]*

6 [7 pts]

In problem DUP, the stream is a sequence of numbers from $[m]$ and the algorithm must say whether some number appears at least twice. You may use: any randomized one-way protocol for Index on m bits with error $\leq 1/3$ needs $\Omega(m)$ bits.

(a, 4 pts) Show that every one-pass streaming algorithm for DUP that succeeds with probability $\geq 2/3$ uses $\Omega(m)$ bits of memory.

Solution. Reduction from Index. Alice holds $x \in \{0, 1\}^m$, Bob holds i. Alice feeds the stream $\{j : x_j = 1\}$ (each once) into the algorithm and sends its memory (S bits) to Bob. Bob continues the run with the single element i and outputs the answer. Correctness: if $x_i = 1$, i already appeared, so i occurs twice ⇒ DUP = yes; if $x_i = 0$, every element appears exactly once ⇒ no. Hence the protocol solves Index with the algorithm's error $\leq 1/3$ and uses S bits, so $S = \Omega(m)$. *[1: Alice's stream; 1: Bob's part + message = memory; 1: both directions; 1: conclusion]*

(b, 1 pt) Why is it fine that the streaming algorithm is randomized?

Solution. Alice and Bob use the algorithm's random bits as shared (public) randomness; Index's $\Omega(m)$ bound holds for randomized protocols with error $1/3$ (and by Newman, public vs. private only changes $O(\log m)$ bits).

(c, 2 pts) Use the same idea to show that computing the *exact* number of distinct elements DE needs $\Omega(m)$ bits.

Solution. Same Alice stream. Bob receives memory state M and makes two copies. On copy 1 he stops immediately: answer $|X|$. On copy 2 he feeds i: answer $|X \cup \{i\}|$. The two are equal iff $x_i = 1$. For a randomized algorithm, first amplify (median of $O(1)$ runs) so each answer is wrong w.p. $\leq 1/6$; union bound ⇒ error $\leq 1/3$. Hence $\Omega(m)$ bits. (Alternatively, Alice sends $|X|$ with $O(\log m)$ extra bits.) *[1: two-copy / extra-count trick; 1: equality ⇔ $x_i = 1$ and error handling]*